Arrays

1. Low-Level Arrays

2. Compact Array

3. Multidimensional Arrays (Matrix)

4. Sparse Martrix

A.1 Low-Level Arrays

In this section we will look at the principles behind low-level arrays. In `C++` we get to work with the real thing: a `C++` array **is** a contiguous block of memory — the very construct that Python's `list` is built on top of in its C implementation. Understanding it explains the performance of every sequence-based structure we will meet.

In this section we will look at the principles behind low-level arrays. In `C++` we get to work with the real thing: a `C++` array **is** a contiguous block of memory — the very construct that Python's `list` is built on top of in its C implementation. Understanding it explains the performance of every sequence-based structure we will meet.

The key to `Python`'s implementation of a `list` is to use a data type called an array common to C, C++, Java, and many other programming languages.

In this section we will look at the principles behind low-level arrays. In `C++` we get to work with the real thing: a `C++` array **is** a contiguous block of memory — the very construct that Python's `list` is built on top of in its C implementation. Understanding it explains the performance of every sequence-based structure we will meet.

The key to `Python`'s implementation of a `list` is to use a data type called an array common to C, C++, Java, and many other programming languages.

The array is very simple and is only capable of **storing one kind of data**. For example, you could have an array of integers or an array of floating point numbers, but you cannot mix the two in a single array.

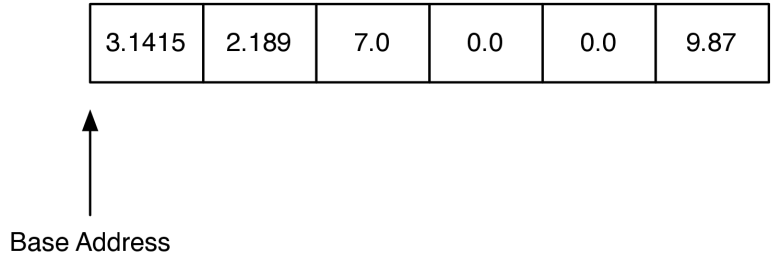
The array only supports two operations: **indexing and assignment to an array index.**

The array only supports two operations: **indexing and assignment to an array index.**

The best way to think about an array is that it is one **continuous block of bytes** in the computer's memory. This block is divided up into -byte chunks where is based on the data type that is stored in the array. Figure below illustrates the idea of an array that is sized to hold six floating point values.

The array only supports two operations: **indexing and assignment to an array index.**

The best way to think about an array is that it is one **continuous block of bytes** in the computer's memory. This block is divided up into -byte chunks where is based on the data type that is stored in the array. Figure below illustrates the idea of an array that is sized to hold six floating point values.



In `C++`, each `double` uses 8 bytes of memory, so an array of six doubles uses a total of 48 bytes. The base address is the location in memory where the array starts. The `sizeof` operator reports the size of any type or object:

In `C++`, each `double` uses 8 bytes of memory, so an array of six doubles uses a total of 48 bytes. The base address is the location in memory where the array starts. The `sizeof` operator reports the size of any type or object:

You have seen addresses before in `Python` for different objects that you have defined. For example: `<__main__.Foo object at 0x5eca30>` shows that the object `Foo` is stored at memory address `0x5eca30`. The address is very important because an array implements the index operator using a very simple calculation:

In `C++`, each `double` uses 8 bytes of memory, so an array of six doubles uses a total of 48 bytes. The base address is the location in memory where the array starts. The `sizeof` operator reports the size of any type or object:

You have seen addresses before in `Python` for different objects that you have defined. For example: `<__main__.Foo object at 0x5eca30>` shows that the object `Foo` is stored at memory address `0x5eca30`. The address is very important because an array implements the index operator using a very simple calculation:

```
item_address = base_address + index * size_of_object
```

For example, suppose that our array starts at location `0x000040`, which is 64 in decimal. To calculate the location of the object at position 4 in the array we simply do the arithmetic:

Clearly this kind of calculation is .

For example, suppose that our array starts at location `0x000040`, which is 64 in decimal. To calculate the location of the object at position 4 in the array we simply do the arithmetic:

Clearly this kind of calculation is .

Of course this comes with some risks:

- First, since the size of an array is fixed, one cannot just add things on to the end of the array indefinitely without some serious consequences.

For example, suppose that our array starts at location `0x000040`, which is 64 in decimal. To calculate the location of the object at position 4 in the array we simply do the arithmetic:

Clearly this kind of calculation is .

Of course this comes with some risks:

- First, since the size of an array is fixed, one cannot just add things on to the end of the array indefinitely without some serious consequences.
- Second, in some languages, like C, the bounds of the array are not even checked, so even though your array has only six elements in it, assigning a value to index 7 will not result in a runtime error!

In [ ]:
```
display_quiz(path+"array1.json", max_width=800)
```

If a low-level array starts at memory address 1024 and each element (e.g., an integer) occupies 4 bytes of memory, what is the address of the element at index 8?

- ○ 1032
- ○ 1056
- ○ 1028
- ○ 1060

As you might imagine this can cause big problems that are hard to track down. In the Linux operating system, accessing a value that is beyond the boundaries of an array will often produce the rather uninformative error message "segmentation fault."

As you might imagine this can cause big problems that are hard to track down. In the Linux operating system, accessing a value that is beyond the boundaries of an array will often produce the rather uninformative error message "segmentation fault."

source:https://www.deviantart.com/froaproa/art/Programming-meme-983116183

The general strategy that a dynamic array (like `C++`'s `vector` or Python's `list`) uses is as follows:

- Allocate a raw array with **extra capacity**, and remember how many slots are actually used.
- When the array fills up, allocate a **bigger** raw array (typically twice the size), copy the elements over, and free the old one.
- Keep indexing O(1) by storing elements contiguously.

The general strategy that a dynamic array (like `C++`'s `vector` or Python's `list`) uses is as follows:

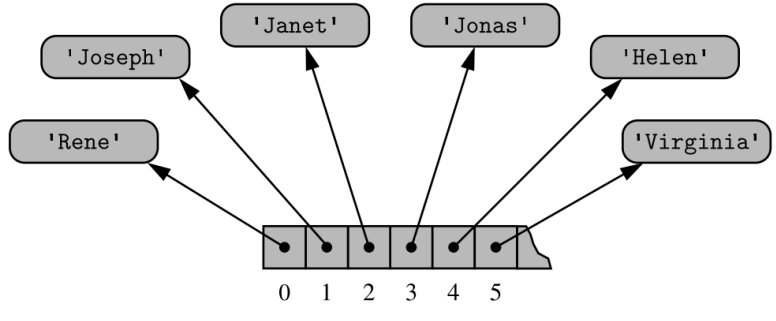
- Allocate a raw array with **extra capacity**, and remember how many slots are actually used.
- When the array fills up, allocate a **bigger** raw array (typically twice the size), copy the elements over, and free the old one.
- Keep indexing O(1) by storing elements contiguously.

- Python uses a strategy called over-allocation to allocate an array with space for more objects than is needed initially.

The general strategy that a dynamic array (like `C++`'s `vector` or Python's `list`) uses is as follows:

- Allocate a raw array with **extra capacity**, and remember how many slots are actually used.
- When the array fills up, allocate a **bigger** raw array (typically twice the size), copy the elements over, and free the old one.
- Keep indexing O(1) by storing elements contiguously.

- Python uses a strategy called over-allocation to allocate an array with space for more objects than is needed initially.



Let's look at how the strategy outlined above works for a very simple implementation. To begin, we will only implement the constructor and the `append()`, `size()` and `isEmpty()` methods. We will call this class `ArrayList`.

Let's look at how the strategy outlined above works for a very simple implementation. To begin, we will only implement the constructor and the `append()`, `size()` and `isEmpty()` methods. We will call this class `ArrayList`.

We will use a `list` to simulate an array:

Let's look at how the strategy outlined above works for a very simple implementation. To begin, we will only implement the constructor and the `append()`, `size()` and `isEmpty()` methods. We will call this class `ArrayList`.

We will use a `list` to simulate an array:

```
class ArrayList {
    public:
        ArrayList(int initialCapacity = 8) {
            maxSize = initialCapacity;
            lastIndex = 0;
            myArray = new int[maxSize];
        }
        ~ArrayList() {
            delete[] myArray;
        }
    private:
        int maxSize;
        int lastIndex;
        int* myArray;   // raw dynamic array
};
```

Let's look at how the strategy outlined above works for a very simple implementation. To begin, we will only implement the constructor and the `append()`, `size()` and `isEmpty()` methods. We will call this class `ArrayList`.

We will use a `list` to simulate an array:

```
class ArrayList {
    public:
        ArrayList(int initialCapacity = 8) {
            maxSize = initialCapacity;
            lastIndex = 0;
            myArray = new int[maxSize];
        }
        ~ArrayList() {
            delete[] myArray;
        }
    private:
        int maxSize;
        int lastIndex;
        int* myArray;   // raw dynamic array
};
```

```
void append(int val) {
    myArray[lastIndex] = val;
    lastIndex++;
}
int size() {
    return lastIndex;
```

```
}
bool isEmpty() {
    return lastIndex == 0;
}
```

Next, let us turn to the index operators. Below is our `C++` implementation for indexing — we overload `operator[]` and check the bounds ourselves, throwing an exception on a bad index:

Next, let us turn to the index operators. Below is our `C++` implementation for indexing — we overload `operator[]` and check the bounds ourselves, throwing an exception on a bad index:

Even languages like C hide that calculation behind a nice array index operator, so in this case the C and the `Python` look very much the same:

Next, let us turn to the index operators. Below is our `C++` implementation for indexing — we overload `operator[]` and check the bounds ourselves, throwing an exception on a bad index:

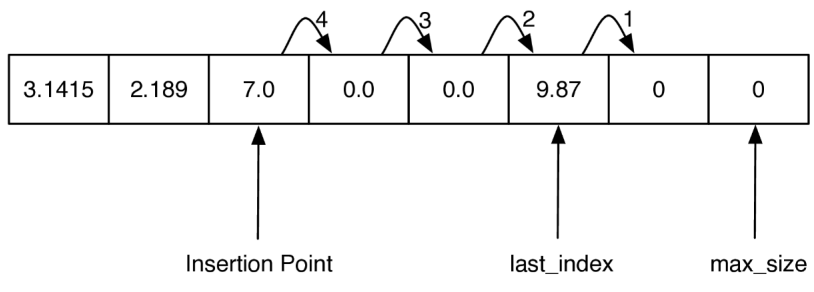
Even languages like C hide that calculation behind a nice array index operator, so in this case the C and the `Python` look very much the same:

```
int& operator[](int idx) {
    if (0 <= idx && idx < lastIndex) {
        return myArray[idx];
    }
    throw out_of_range("index out of bounds");
}
```

Returning a **reference** (`int&`) makes both reading `a[i]` and assigning `a[i] = val` work with a single operator — the compiler picks the right use automatically.

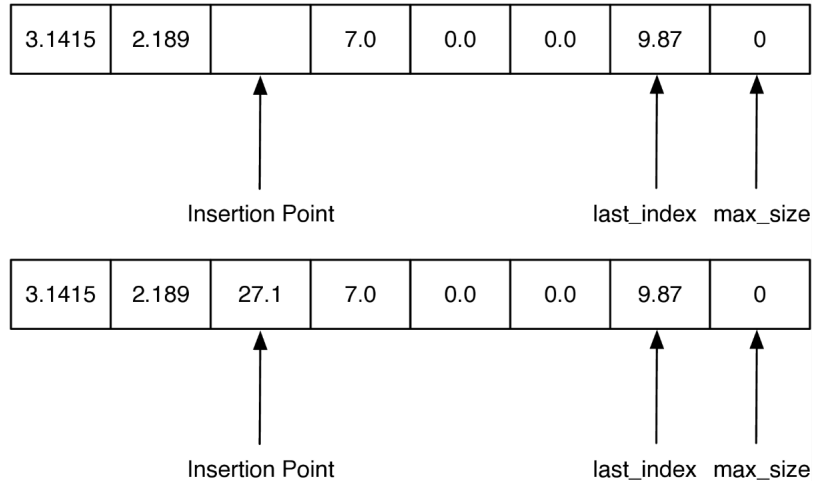
Finally, let's take a look at one of the more expensive list operations, `insertion()`. When we insert an item into an `ArrayList` we will need to first shift everything in the list at the insertion point and beyond ahead by one index position in order to make room for the item we are inserting. The process is illustrated in Figure below:

Finally, let's take a look at one of the more expensive list operations, `insertion()`. When we insert an item into an `ArrayList` we will need to first shift everything in the list at the insertion point and beyond ahead by one index position in order to make room for the item we are inserting. The process is illustrated in Figure below:



The key to implementing insert correctly is to realize that as you are shifting values in the array you do not want to overwrite any important data. **The way to do this is to work from the end of the list back toward the insertion point, copying data forward.**

The key to implementing insert correctly is to realize that as you are shifting values in the array you do not want to overwrite any important data. **The way to do this is to work from the end of the list back toward the insertion point, copying data forward.**



Our implementation of insert is shown below. Note how the range is set up on line to ensure that we are copying existing data into the unused part of the array first, and then subsequent values are copied over old values that have already been shifted.

Our implementation of insert is shown below. Note how the range is set up on line to ensure that we are copying existing data into the unused part of the array first, and then subsequent values are copied over old values that have already been shifted.

```
void insert(int idx, int val) {
    for (int i = lastIndex; i > idx; i--) {
        myArray[i] = myArray[i - 1];
    }
    myArray[idx] = val;
    lastIndex++;
}
```

Our implementation of insert is shown below. Note how the range is set up on line to ensure that we are copying existing data into the unused part of the array first, and then subsequent values are copied over old values that have already been shifted.

```
void insert(int idx, int val) {
    for (int i = lastIndex; i > idx; i--) {
        myArray[i] = myArray[i - 1];
    }
    myArray[idx] = val;
    lastIndex++;
}
```

If the loop had started at the insertion point and copied that value to the next larger index position in the array, the old value would have been lost forever!

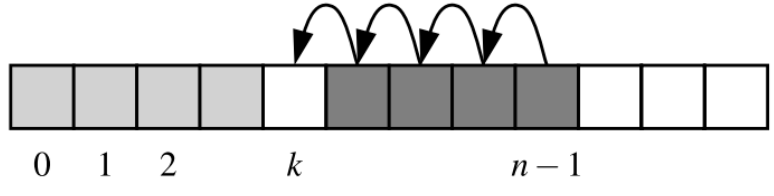
The performance of the insert is since in the worst case we want to insert something at index 0 and we have to shift the entire array forward by one. On average we will only need to shift half of the array, but this is still .

The performance of the insert is since in the worst case we want to insert something at index 0 and we have to shift the entire array forward by one. On average we will only need to shift half of the array, but this is still .

The class offers another method, named `remove()`, that allows the caller to specify the value that should be removed (not the index at which it resides). Formally, it removes only the first occurrence of such a value from an array, or raises an error if no such value is found.

The performance of the insert is since in the worst case we want to insert something at index 0 and we have to shift the entire array forward by one. On average we will only need to shift half of the array, but this is still .

The class offers another method, named `remove()`, that allows the caller to specify the value that should be removed (not the index at which it resides). Formally, it removes only the first occurrence of such a value from an array, or raises an error if no such value is found.



```
void remove(int val) {
    for (int i = 0; i < lastIndex; i++) {
        if (myArray[i] == val) {
            for (int j = i; j < lastIndex - 1; j++) {
                myArray[j] = myArray[j + 1];
            }
            lastIndex--;
            return;
        }
    }
    throw invalid_argument("value not found");
}
```

```
void remove(int val) {
    for (int i = 0; i < lastIndex; i++) {
        if (myArray[i] == val) {
            for (int j = i; j < lastIndex - 1; j++) {
                myArray[j] = myArray[j + 1];
            }
            lastIndex--;
            return;
        }
    }
    throw invalid_argument("value not found");
}
```

One part of the process searches from the beginning until finding the value at index k, while the rest iterates from k to the end in order to shift elements leftward. The complexity is again .

The complete class (with `erase()`, growth handling, and printing) is collected in `pythonds3/cppds/basic/`. Here is the whole thing in action:

In [2]:
```cpp
#include <iostream>
#include "dscpp/arraylist.hpp"   // the complete class of this section
using namespace std;

int main() {
    ArrayList myArray;
    myArray.append(31); myArray.append(77);
    myArray.append(17); myArray.append(93);
    myArray.display();
    cout << myArray[3] << " " << myArray.size() << endl;

    myArray.insert(3, 20);
    myArray.remove(31);
    myArray.display();

    myArray.erase(2);
    myArray[0] = 50;
    myArray.display();
    return 0;
}
```

31 77 17 93

93 4

77 17 20 93

50 17 93

In [ ]:
```
display_quiz(path+"array2.json", max_width=800)
```

In C++, an array of pointers (for example, Animal* zoo[10]) is considered a referential array. Why?

- ○ Because it stores the actual bits of each object directly in the array cells.
- ○ Because each cell stores a memory address that points to an object living elsewhere in memory.
- ○ Because it only allows elements of exactly the same size to be referenced.
- ○ Because pointer arrays are automatically allocated on the heap.

A.2 Compact Array

Assume that we have a collection of names such as `{"Rene", "Joseph", "Janet", "Jonas", "Helen", "Virginia", ...}`. In a referential layout the array cells hold **pointers** to string objects living elsewhere; each cell has the same fixed size no matter how long the name is.

Assume that we have a collection of names such as `{"Rene", "Joseph", "Janet", "Jonas", "Helen", "Virginia", ...}`. In a referential layout the array cells hold **pointers** to string objects living elsewhere; each cell has the same fixed size no matter how long the name is.

To represent such a list with an array, `Python` must adhere to the requirement that each cell of the array use the same number of bytes. Yet the elements are strings, and strings naturally have different lengths!

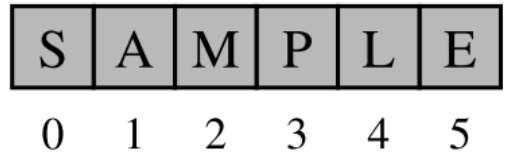
Assume that we have a collection of names such as `{"Rene", "Joseph", "Janet", "Jonas", "Helen", "Virginia", ...}`. In a referential layout the array cells hold **pointers** to string objects living elsewhere; each cell has the same fixed size no matter how long the name is.

To represent such a list with an array, `Python` must adhere to the requirement that each cell of the array use the same number of bytes. Yet the elements are strings, and strings naturally have different lengths!

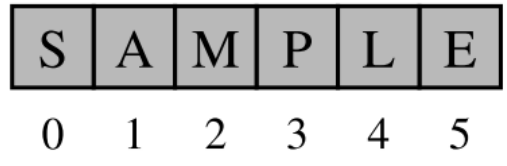
Instead, `Python` represents a list or tuple instance using an internal storage mechanism of an **array of object references.** Although the relative size of the individual elements may vary, the number of bits used to store the memory address of each element is fixed (e.g., 64-bits per address). In this way, `Python` can support constant-time access!

On the other hand, `strings` are represented using an array of characters (not an array of references). We will refer to this more direct representation as a compact array because the array is storing the bits that represent the primary data (characters, in the case of strings)!

On the other hand, `strings` are represented using an array of characters (not an array of references). We will refer to this more direct representation as a compact array because the array is storing the bits that represent the primary data (characters, in the case of strings)!



On the other hand, `strings` are represented using an array of characters (not an array of references). We will refer to this more direct representation as a compact array because the array is storing the bits that represent the primary data (characters, in the case of strings)!



The overall memory usage will be much lower for a compact structure because there is no overhead devoted to the explicit storage of the sequence of memory references (in addition to the primary data)!

A referential structure will typically use 8 bytes (64 bits) for the memory address stored in the array, on top of whatever memory the object itself uses! In `C++`, an ordinary array **is already compact**: the element type determines exactly how many bytes each slot occupies.

A referential structure will typically use 8 bytes (64 bits) for the memory address stored in the array, on top of whatever memory the object itself uses! In `C++`, an ordinary array **is already compact**: the element type determines exactly how many bytes each slot occupies.

In [3]:
```
using namespace std;
int primes[] = {2, 3, 5, 7, 11, 13, 17, 19};

cout << sizeof(primes[0]) << " bytes per element" << endl;
cout << sizeof(primes) << " bytes in total" << endl;
cout << sizeof(primes) / sizeof(primes[0]) << " elements" << endl;

string* names[3];   // a referential array: 3 pointers, 8 bytes each
cout << sizeof(names) << " bytes for three pointers" << endl;
```

4 bytes per element

32 bytes in total

8 elements

24 bytes for three pointers

A referential structure will typically use 8 bytes (64 bits) for the memory address stored in the array, on top of whatever memory the object itself uses! In `C++`, an ordinary array **is already compact**: the element type determines exactly how many bytes each slot occupies.

In [3]:
```
using namespace std;
int primes[] = {2, 3, 5, 7, 11, 13, 17, 19};

cout << sizeof(primes[0]) << " bytes per element" << endl;
cout << sizeof(primes) << " bytes in total" << endl;
cout << sizeof(primes) / sizeof(primes[0]) << " elements" << endl;

string* names[3];   // a referential array: 3 pointers, 8 bytes each
cout << sizeof(names) << " bytes for three pointers" << endl;
```

4 bytes per element

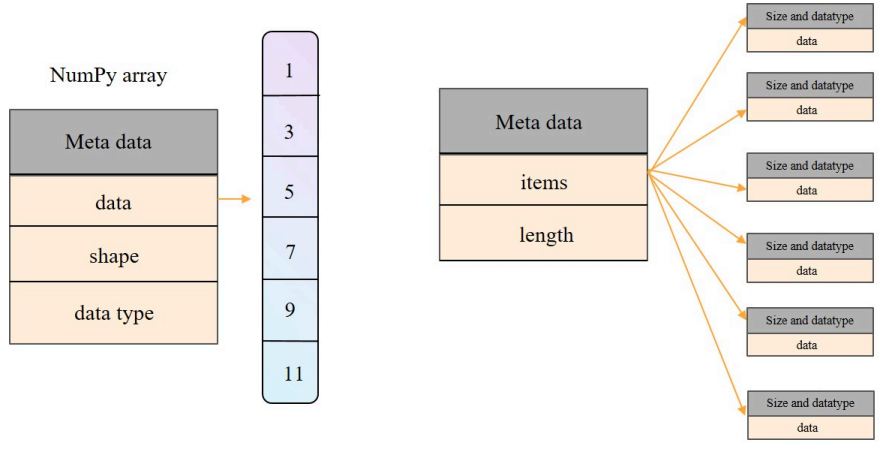
32 bytes in total

8 elements

24 bytes for three pointers

The element type plays the role of NumPy's type code: declaring `int` (4 bytes) versus `double` (8 bytes) versus `char` (1 byte) tells the **compiler** precisely how many bits are needed per element of the array — no interpreter needed at run time.

The element type plays the role of NumPy's type code: declaring `int` (4 bytes) versus `double` (8 bytes) versus `char` (1 byte) tells the **compiler** precisely how many bits are needed per element of the array — no interpreter needed at run time.



In [ ]:
```
display_quiz(path+"array3.json", max_width=800)
```

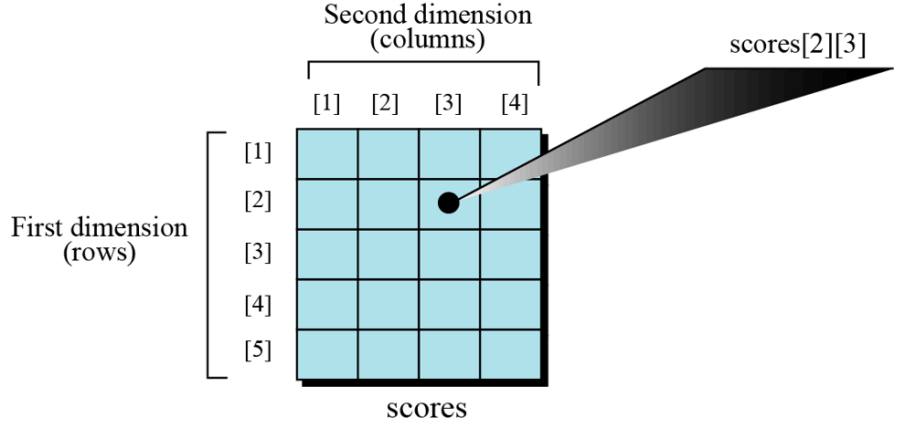
When using a C++ vector, which of the following operations has a time complexity of $O(n)$ due to the need to shift elements?

- ○ vec[i] = value
- ○ vec.pop_back()
- ○ vec.push_back(value)
- ○ vec.insert(vec.begin(), value)

A.3 Multidimensional Arrays

The arrays discussed so far are known as one-dimensional arrays because the data is organized linearly in only one direction. Many applications require that data be stored in more than one dimension. One common example is a table, which is an array that consists of rows and columns. This is commonly called a two-dimensional array (Matrix):

The arrays discussed so far are known as one-dimensional arrays because the data is organized linearly in only one direction. Many applications require that data be stored in more than one dimension. One common example is a table, which is an array that consists of rows and columns. This is commonly called a two-dimensional array (Matrix):



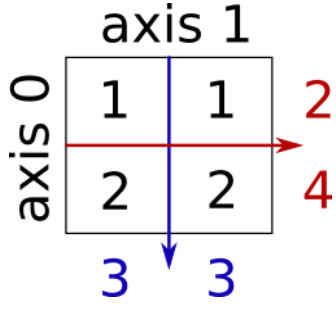
The array holds the scores of students in a class. There are five students in the class and each student has four different scores for four subjects. The variable `scores[2][3]` show the score of the second student in the third quiz.

The array holds the scores of students in a class. There are five students in the class and each student has four different scores for four subjects. The variable `scores[2][3]` show the score of the second student in the third quiz.

Arranging the scores in a two-dimensional array can help us to find the average of scores for each student (the average over the row values) and find the average for each subject (the average over the column values) and so on. Note that multidimensional arrays — arrays with more than two dimensions — are also possible.

The array holds the scores of students in a class. There are five students in the class and each student has four different scores for four subjects. The variable `scores[2][3]` show the score of the second student in the third quiz.

Arranging the scores in a two-dimensional array can help us to find the average of scores for each student (the average over the row values) and find the average for each subject (the average over the column values) and so on. Note that multidimensional arrays — arrays with more than two dimensions — are also possible.



source: https://scipy-lectures.org/intro/numpy/operations.html

In [4]:
```
using namespace std;
// a matrix: initialize a 2D array with nested braces
int M[2][2] = {{1, 1}, {2, 2}};

cout << "M has " << sizeof(M) / sizeof(M[0]) << " rows and "
     << sizeof(M[0]) / sizeof(M[0][0]) << " columns, "
     << sizeof(M) << " bytes in total" << endl;
```

M has 2 rows and 2 columns, 16 bytes in total

In [4]:
```cpp
using namespace std;
// a matrix: initialize a 2D array with nested braces
int M[2][2] = {{1, 1}, {2, 2}};

cout << "M has " << sizeof(M) / sizeof(M[0]) << " rows and "
     << sizeof(M[0]) / sizeof(M[0][0]) << " columns, "
     << sizeof(M) << " bytes in total" << endl;
```

M has 2 rows and 2 columns, 16 bytes in total

Note the slicing of array works the same way as `Python` `list`!

In [5]:
```cpp
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
for (int j = 0; j < 2; j++) {
    cout << M[0][j] << " ";   // row 0
}
cout << endl;
```

```
1 1
```

In [5]:
```cpp
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
for (int j = 0; j < 2; j++) {
    cout << M[0][j] << " ";   // row 0
}
cout << endl;
```

```
1 1
```

In [6]:
```cpp
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
for (int i = 0; i < 2; i++) {
    cout << M[i][0] << " ";   // column 0
}
cout << endl;
```

```
1 2
```

In [5]:
```
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
for (int j = 0; j < 2; j++) {
    cout << M[0][j] << " ";   // row 0
}
cout << endl;
```

```
1 1
```

In [6]:
```
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
for (int i = 0; i < 2; i++) {
    cout << M[i][0] << " ";   // column 0
}
cout << endl;
```

```
1 2
```

In [7]:
```
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
cout << M[1][1] << endl;   // row 1, column 1
```

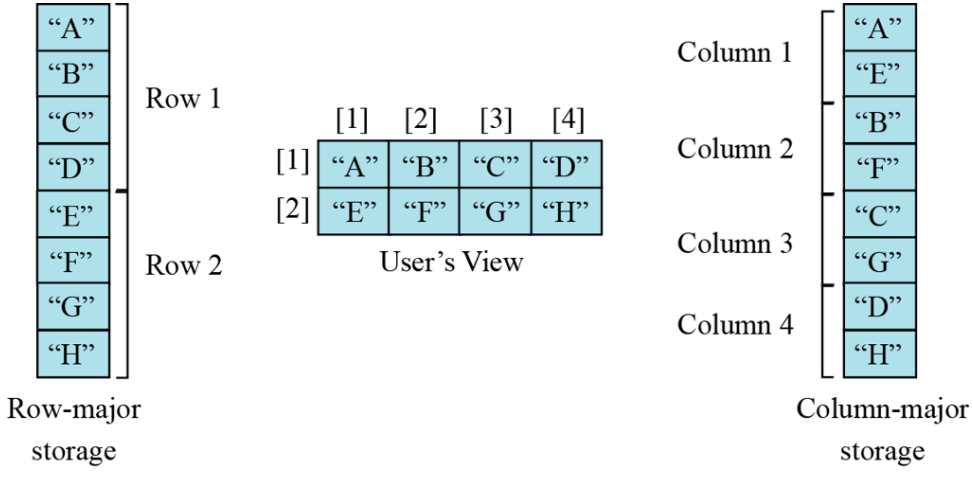
```
2
```

A.3.1 Memory layout

The indexes in a one-dimensional array directly define the relative positions of the element in actual memory. A two-dimensional array, however, represents rows and columns. How each element is stored in memory depends on the computer!

The indexes in a one-dimensional array directly define the relative positions of the element in actual memory. A two-dimensional array, however, represents rows and columns. How each element is stored in memory depends on the computer!

Most computers use **row-major** storage, in which an entire row of an array is stored in memory before the next row. However, a computer may store the array using **column-major** storage, in which the entire column is stored before the next column. The following figure shows a two-dimensional array and how it is stored in memory using row-major or column-major storage. Row-major storage is more common.


"A"
"B"
"C"
"D"
"E"
"F"
"G"
"H"
Row 1
Row 2
Row-major storage
[1] [2] [3] [4]
[1] "A" "B" "C" "D"
[2] "E" "F" "G" "H"
User's View
Column 1
Column 2
Column 3
Column 4
"A"
"E"
"B"
"F"
"C"
"G"
"D"
"H"
Column-major storage

`C++` two-dimensional arrays are **row-major by definition** — the language standard guarantees that `M[0][0], M[0][1], M[1][0], M[1][1]` sit consecutively in memory. We can see it directly by printing addresses:

`C++` two-dimensional arrays are **row-major by definition** — the language standard guarantees that `M[0][0], M[0][1], M[1][0], M[1][1]` sit consecutively in memory. We can see it directly by printing addresses:

In [8]:
```cpp
using namespace std;
int M[2][2] = {{1, 1}, {2, 2}};
cout << &M[0][0] << " " << &M[0][1] << endl;   // 4 bytes apart
cout << &M[1][0] << " " << &M[1][1] << endl;   // next row follows immediately
```

0x7ffee0f64680 0x7ffee0f64684

0x7ffee0f64688 0x7ffee0f6468c

`C++` has no built-in column-major layout (that is Fortran's convention). If an algorithm needs it, you store the matrix in a 1D array and do the index arithmetic yourself — note the formula swaps the roles of rows and columns:

`C++` has no built-in column-major layout (that is Fortran's convention). If an algorithm needs it, you store the matrix in a 1D array and do the index arithmetic yourself — note the formula swaps the roles of rows and columns:

In [9]:
```cpp
using namespace std;
// column-major storage of a 2x2 matrix in a flat array
int flat[4];
int rows = 2, cols = 2;
int M[2][2] = {{1, 1}, {2, 2}};

for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        flat[j * rows + i] = M[i][j];   // column-major: j*Rows + i

for (int k = 0; k < 4; k++) cout << flat[k] << " ";
cout << endl;
```

```
1 2 1 2
```

> *Exercise 1: We have stored the two-dimensional array students in the memory. The array is 100 × 4 (100 rows and 4 columns). Show the address of the element* `students[5][3]` *assuming that the element* `student[0][0]` *is stored in the memory location with address 0 and each element occupies only one memory location. The computer uses row-major storage.*

Exercise 1: We have stored the two-dimensional array students in the memory. The array is 100 × 4 (100 rows and 4 columns). Show the address of the element `students[5][3]` assuming that the element `student[0][0]` is stored in the memory location with address 0 and each element occupies only one memory location. The computer uses row-major storage.

In [10]:
```
using namespace std;
int student[100][4];
for (int i = 0; i < 100; i++)
    for (int j = 0; j < 4; j++)
        student[i][j] = i * 4 + j;   // fill with recognizable values

int* s = &student[0][0];   // view the 2D array as a flat 1D array
cout << student[5][3] << endl;
```

23

> *Exercise 1: We have stored the two-dimensional array students in the memory. The array is 100 × 4 (100 rows and 4 columns). Show the address of the element* `students[5][3]` *assuming that the element* `student[0][0]` *is stored in the memory location with address 0 and each element occupies only one memory location. The computer uses row-major storage.*

In [10]:
```cpp
using namespace std;
int student[100][4];
for (int i = 0; i < 100; i++)
    for (int j = 0; j < 4; j++)
        student[i][j] = i * 4 + j;   // fill with recognizable values

int* s = &student[0][0];   // view the 2D array as a flat 1D array
cout << student[5][3] << endl;
```

23

In [11]:
```cpp
#include <cassert>
using namespace std;

int main() {
    int student[100][4];
    for (int i = 0; i < 100; i++)
```

```
        for (int j = 0; j < 4; j++) student[i][j] = i * 4 + j;
    int* s = &student[0][0];
    // Replace ? with your answer
    assert(student[5][3] == s[?]);
    cout << "Pass" << endl;
    return 0;
}
```

```
In file included from /usr/include/c++/13/cassert:44,
                 from /tmp/tmpikzgk11a.cpp:3:
/tmp/tmpikzgk11a.cpp: In function ‘int main()’:
/tmp/tmpikzgk11a.cpp:12:31: error: expected primary-expression before
‘?’ token
   12 |     assert(student[5][3] == s[?]);
      |                               ^
/tmp/tmpikzgk11a.cpp:12:32: error: expected primary-expression before
‘]’ token
   12 |     assert(student[5][3] == s[?]);
    ... (truncated; run the cell to see the full message)
```

We can use the following formula to find the location of an element, assuming each element occupies one memory location:

$$y = x + Cols \times i + j$$

where `x` defines the start address, Cols defines the number of columns in the array, `i` defines the row number of the element, `j` defines the column number of the element, and `y` is the address we are looking for!

In [ ]:
```
display_quiz(path+"array4.json", max_width=800)
```

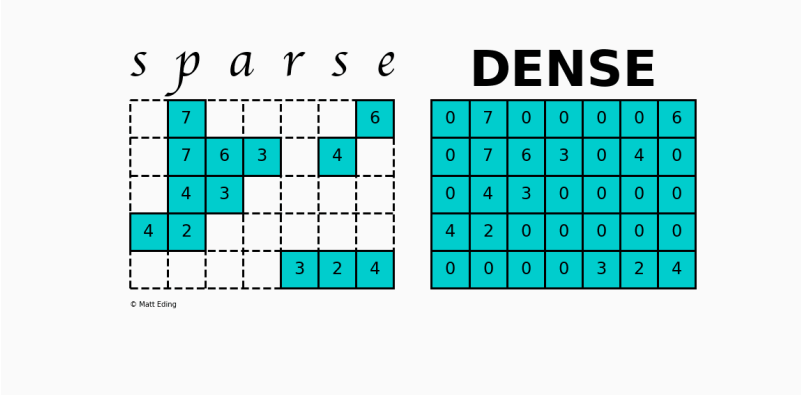
In a 2D array (Matrix) with 5 rows and 10 columns stored in Row-Major order, which formula represents the 1D index of the element at row i and column j?

- ○ `i * 5 + j`
- ○ `j * 5 + i`
- ○ `i + j`
- ○ `i * 10 + j`

A.4 Sparse Martrix

A sparse matrix is a matrix that has a value of 0 for most elements. If the ratio of Number of Non-Zero (NNZ) elements to the size is less than 0.05, the matrix is sparse!

A sparse matrix is a matrix that has a value of 0 for most elements. If the ratio of Number of Non-Zero (NNZ) elements to the size is less than 0.05, the matrix is sparse!



source: https://matteding.github.io/2019/04/25/sparse-matrices/

Storing information about all the 0 elements is inefficient, so we will assume unspecified elements to be 0. Using this scheme, sparse matrices can perform faster operations and use less memory than its corresponding dense matrix representation.

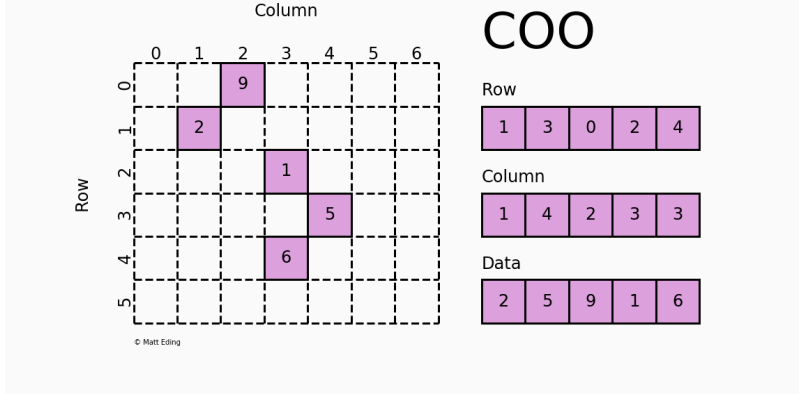
Storing information about all the 0 elements is inefficient, so we will assume unspecified elements to be 0. Using this scheme, sparse matrices can perform faster operations and use less memory than its corresponding dense matrix representation.

Different sparse formats have their strengths and weaknesses. A good starting point is looking at formats that are efficient for constructing these matrices. Typically you would start with one of these forms and then convert to another when ready to do calculations.

A.4.1 Coordinate Matrix

Perhaps the simplest sparse format to understand is the COOrdinate (COO) format. This variant uses three subarrays to store the element values and their coordinate positions.

Perhaps the simplest sparse format to understand is the COOrdinate (COO) format. This variant uses three subarrays to store the element values and their coordinate positions.



source: https://matteding.github.io/2019/04/25/sparse-matrices/

The savings on memory consumption is quite substantial as the matrix size increases. Managing data in a sparse structure is a fixed cost unlike the case for dense matrices.

The savings on memory consumption is quite substantial as the matrix size increases. Managing data in a sparse structure is a fixed cost unlike the case for dense matrices.

The overhead incurred from needing to manage the subarrays is becomes negligible as data grows making it a great choice for some datasets!

A.4.2 Dictionary of Keys Matrix

Dictionary Of Keys (DOK) is very much like COO except that it uses a **map with (row, column) pairs as keys** to store coordinate-data information as key-value pairs. Using a hash-based container, identifying values at any given location has constant lookup time!

Dictionary Of Keys (DOK) is very much like COO except that it uses a **map with (row, column) pairs as keys** to store coordinate-data information as key-value pairs. Using a hash-based container, identifying values at any given location has constant lookup time!

Use this format if you need the functionality that come with builtin dictionaries, but be mindful that hash tables hog more memory than arrays.

Dictionary Of Keys (DOK) is very much like COO except that it uses a **map with (row, column) pairs as keys** to store coordinate-data information as key-value pairs. Using a hash-based container, identifying values at any given location has constant lookup time!

Use this format if you need the functionality that come with builtin dictionaries, but be mindful that hash tables hog more memory than arrays.

The `SparseMatrix` class uses a dictionary to store non-zero elements, with keys as `(row, column)` tuples and values as the non-zero elements.

```
class SparseMatrix {
    public:
        SparseMatrix() {}

        // read access: return the stored value, or 0 for unset positions
        double operator()(size_t i, size_t j) const {
            auto it = data.find({i, j});
            if (it != data.end()) {
                return it->second;
            }
            return 0.0;
        }
    private:
        map<pair<size_t, size_t>, double> data;
};
```

```
class SparseMatrix {
    public:
        SparseMatrix() {}

        // read access: return the stored value, or 0 for unset positions
        double operator()(size_t i, size_t j) const {
            auto it = data.find({i, j});
            if (it != data.end()) {
                return it->second;
            }
            return 0.0;
        }
    private:
        map<pair<size_t, size_t>, double> data;
};
```

```
// write access: inserts the position if absent
double& operator()(size_t i, size_t j) {
    return data[{i, j}];
}

size_t nnz() const {
    return data.size();   // number of non-zeros
}
```

We can easily implement relational operations as follows:

We can easily implement relational operations as follows:

```
bool operator==(const SparseMatrix& other) const {
    return data == other.data;
}

bool operator!=(const SparseMatrix& other) const {
    return !(*this == other);
}
```

We can easily implement relational operations as follows:

```
bool operator==(const SparseMatrix& other) const {
    return data == other.data;
}

bool operator!=(const SparseMatrix& other) const {
    return !(*this == other);
}
```

```
// how sparse is the matrix, given its logical dimensions?
double sparsity(size_t rows, size_t cols) const {
    return 1.0 - double(data.size()) / double(rows * cols);
}
```

The arithmetic operation is slightly more involved — every coordinate present in either operand must appear in the result:

The arithmetic operation is slightly more involved — every coordinate present in either operand must appear in the result:

```
SparseMatrix operator+(const SparseMatrix& other) const {
    SparseMatrix result;
    for (const auto& item : data) {
        result.data[item.first] = item.second + other(item.first.first,
item.first.second);
    }
    for (const auto& item : other.data) {
        if (data.find(item.first) == data.end()) {
            result.data[item.first] = item.second;
        }
    }
    return result;
}
```

Multiplication only needs to visit the non-zero entries of both operands — this is where the sparse representation really pays off:

Multiplication only needs to visit the non-zero entries of both operands — this is where the sparse representation really pays off:

```
SparseMatrix operator*(const SparseMatrix& other) const {
    SparseMatrix result;
    for (const auto& item1 : data) {
        for (const auto& item2 : other.data) {
            if (item1.first.second == item2.first.first) {
                result(item1.first.first, item2.first.second)
                    += item1.second * item2.second;
            }
        }
    }
    return result;
}
```

The conversion from dense matrix can be written as follows:

The conversion from dense matrix can be written as follows:

```
void fromDenseMatrix(const vector<vector<double>>& matrix) {
    for (size_t i = 0; i < matrix.size(); ++i) {
        for (size_t j = 0; j < matrix[i].size(); ++j) {
            if (matrix[i][j] != 0) {
                data[{i, j}] = matrix[i][j];
            }
        }
    }
}
```

We can play the class as follows:

We can play the class as follows:

The complete class lives in `pythonds3/cppds/basic/sparse.cpp`. Here it is in action:

In [12]:

```cpp
#include <iostream>
#include "dscpp/sparsematrix.hpp"
using namespace std;

int main() {
    vector<vector<double>> denseMatrix = {{1, 0, 0}, {0, 2, 0}, {0, 0, 3}};
    SparseMatrix sparseMatrix;
    sparseMatrix.fromDenseMatrix(denseMatrix);
    cout << sparseMatrix << endl;

    SparseMatrix matrix1({{{0, 1}, 1}, {{1, 1}, 2}, {{2, 2}, 3}});
    SparseMatrix matrix2({{{1, 1}, 3}, {{2, 2}, 4}});

    cout << matrix1 + matrix2 << endl;
    cout << matrix1 - matrix2 << endl;
    cout << matrix1 * matrix2 << endl;
    return 0;
}
```

(0, 0):

1 (1, 1): 2 (2, 2): 3

(0, 1): 1 (1, 1): 5 (2, 2): 7

(0, 1): 1 (1, 1): -1 (2, 2): -1

(0, 1): 3 (1, 1): 6 (2, 2): 12

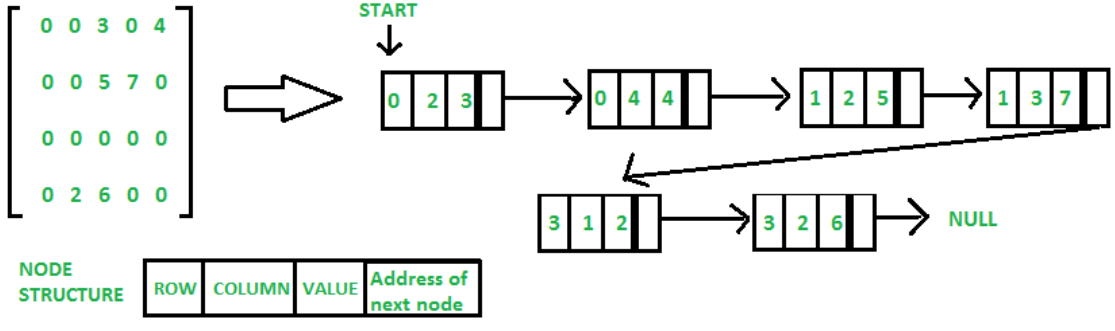
Each printed line lists the non-zero entries as `(row, column): value` pairs — everything absent is an implicit zero.

In [ ]:
```
display_quiz(path+"array5.json", max_width=800)
```

What is the primary reason for using a specialized Sparse Matrix class instead of a standard 2D array?

- ○ To make the matrix operations like addition and multiplication slower but more accurate.
- ○ To allow the matrix to store different data types like strings and integers together.
- ○ To save memory and computation time when the majority of the elements in the matrix are zero.
- ○ Because standard 2D arrays cannot have more than 100 rows.

A.4.3 Linear list



source: https://www.geeksforgeeks.org/sparse-matrix-representation/

In [ ]:

```
from jupytercards import display_flashcards
fpath = "flashcards/"
display_flashcards(fpath + 'ch3.json')
```

Next

>

References

1. Textbook: *Problem Solving with Algorithms and Data Structures using C++* (cppds) — https://runestone.academy/ns/books/published/cppds/index.html
2. Forouzan, *Foundations of Computer Science*, Ch 11 (Data Organization)